

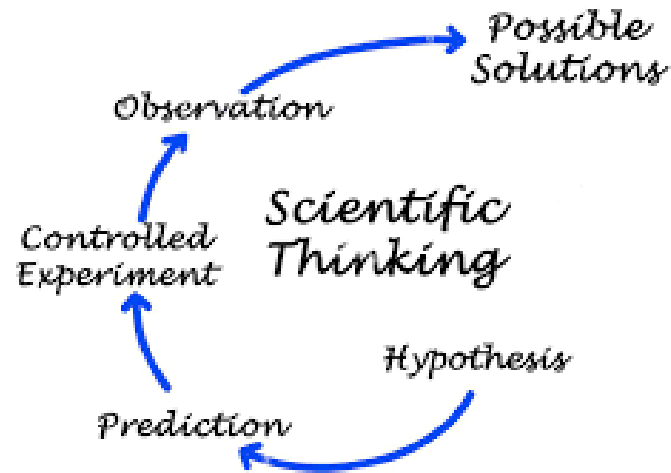


Methods of Scientific Research

Course code : HUM 303

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Development and Methods of Scientific thinking

➤ What Is Scientific Thinking?

➤ Science assumes the natural world is:

- **Consistent.**
- **Predictable.**

➤ The goals of science are:

- **To discover patterns in nature.**
- **To use the knowledge to predict.**



➤ What Is Scientific Inquiry?

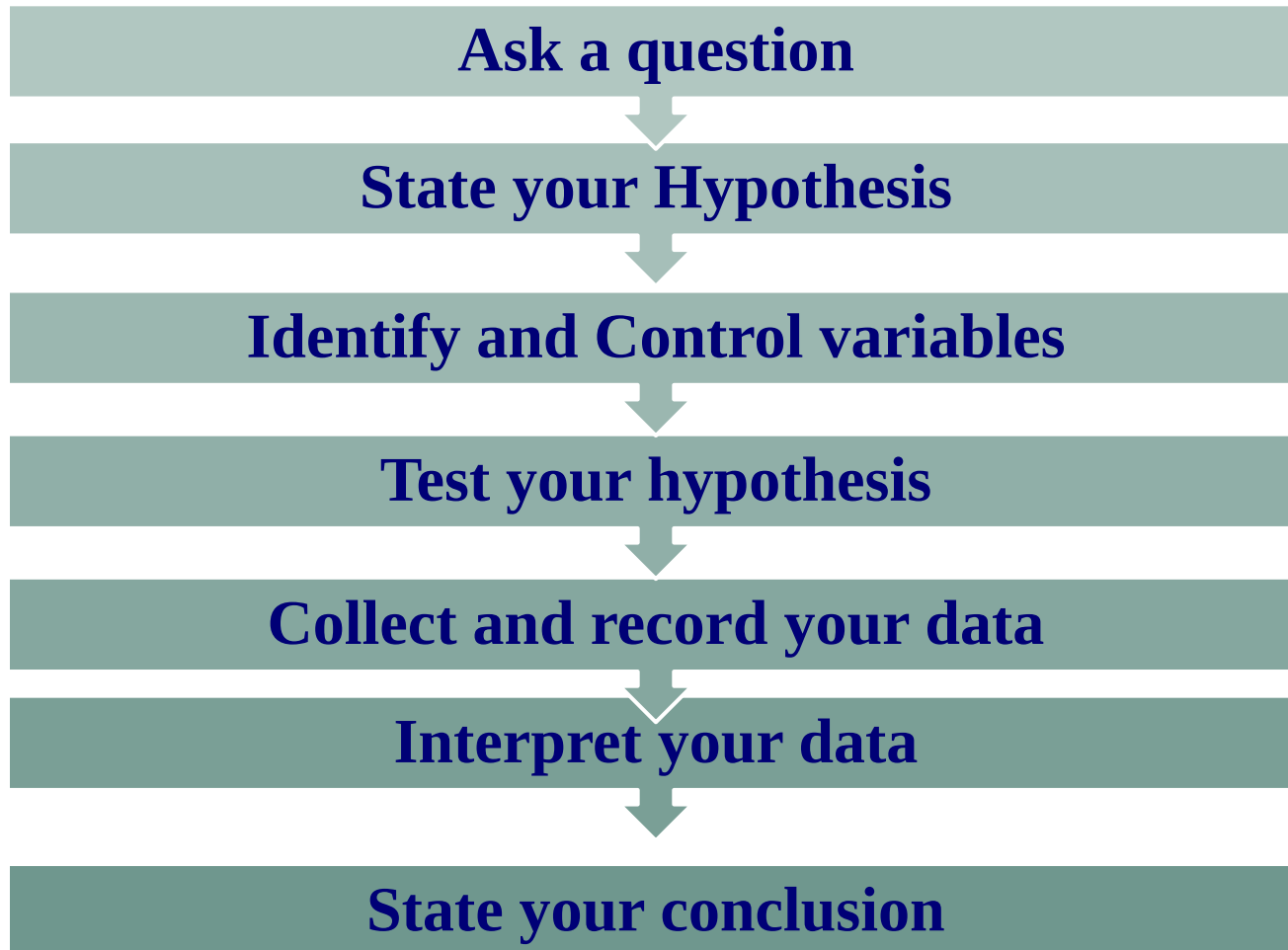
➤ An idea can become a:

- **Hypothesis: Tentative or untested explanation, educated guess.**
- **Theory: Tested, confirmed, supported hypothesis.**

➤ **Scientists use the scientific method, a refinement of every thinking.**

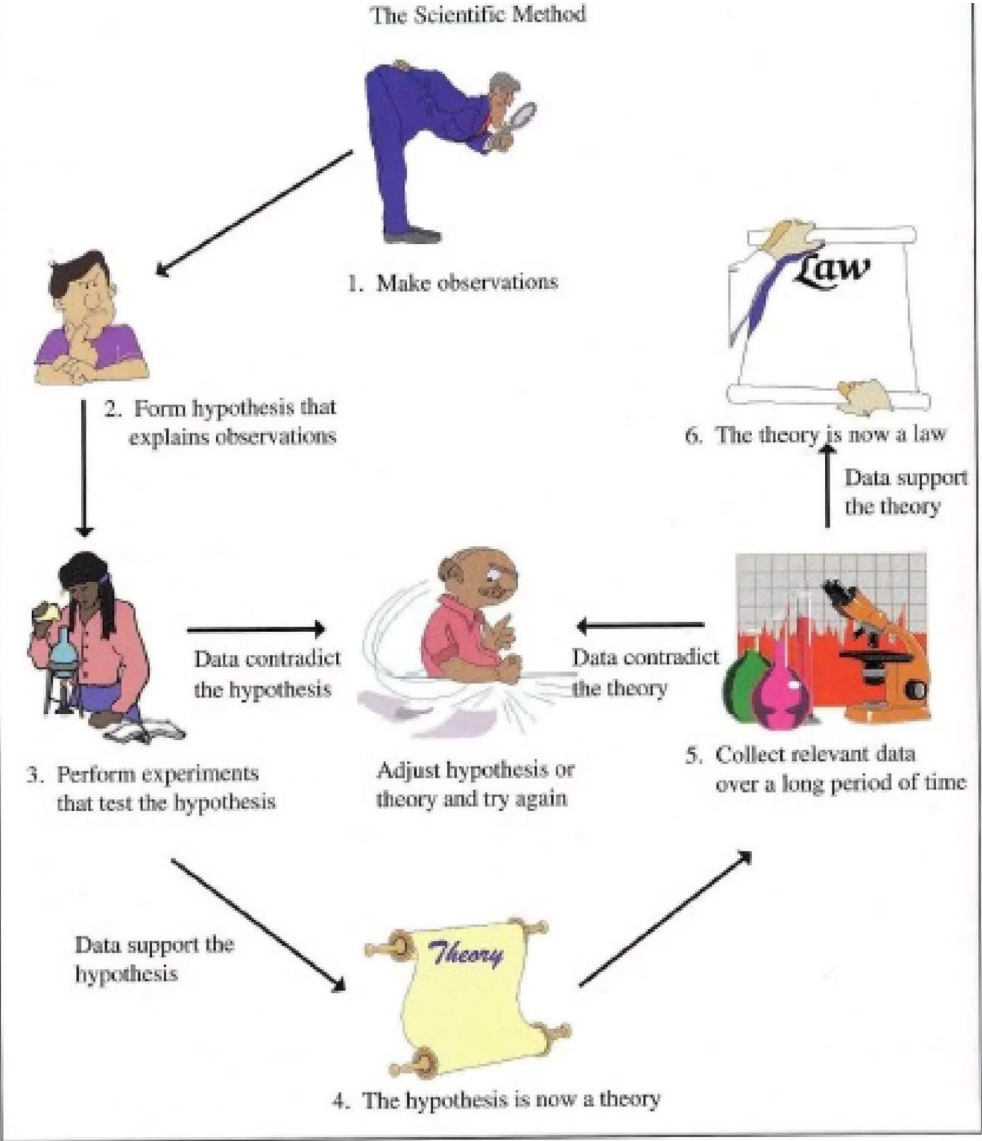
Step one	• Ask a question or Identify a problem
Step two	• Increase knowledge by observation
Step three	• Make a scientific hypothesis
Step four	• Experiment
Step five	• Analyze data
Step six	• Conclusion
Step seven	• Repeat
Step eight	• Scientific theory
Step nine	• Communication

➤ **Another Steps for the Scientific Method**



➤ Example

- **Question:** What material is best for keeping heat in water?
- **Hypothesis:** A glass jar will keep it warmer.
- **Control Variables:** Do the test in different types of jar.
- **Test Your Hypothesis:** Measure the temperature of the different jars.
- **Collect Data:** Write down the measurements.
- **Interpret:** Ask yourself if temperatures really differ.
- **State Your Conclusion:** Tell whether or not your data supported your hypothesis.



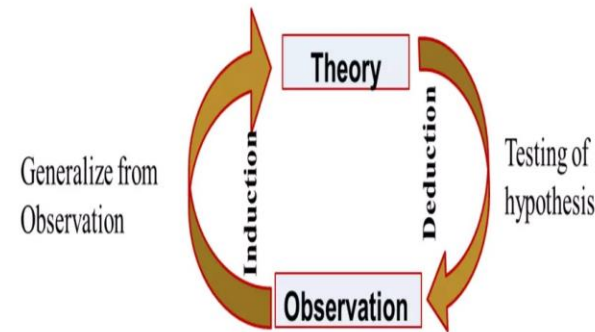
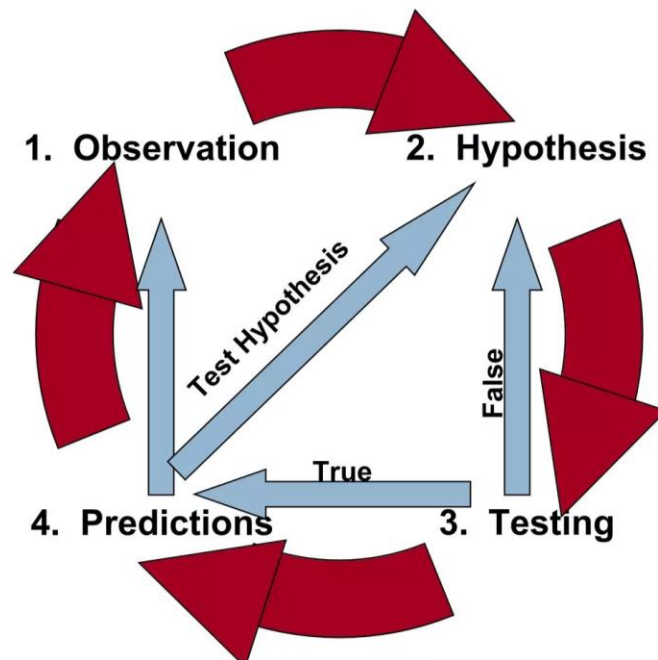
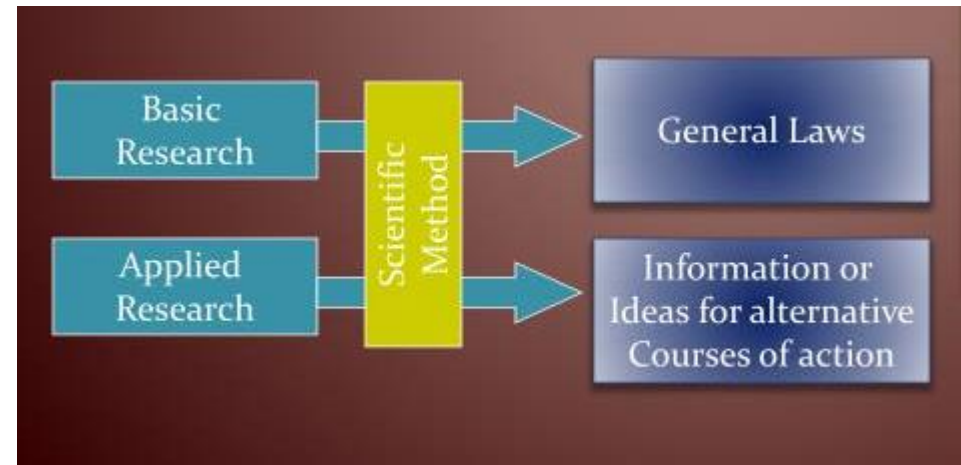


Figure: The cycle of Scientific Research

If the experiments prove the hypothesis to be true, it becomes a theory or law of nature. If the experiments prove the hypothesis to be false, the hypothesis must be rejected or modified. The scientific method used properly should give us predictive power (to understand phenomena that have not been tested).

➤ STEPS OF SCIENTIFIC METHODS

- **Problem/Observation.**
- **Question/Research.**
- **Formulate a Hypothesis.**
- **Experiment.**
- **Collect and Analyze Results.**
- **Conclusion.**
- **Communicate the Results.**



➤ Understanding of Methodology and Methods:

➤ **Methodology** refers to more than a simple set of methods; it relates to the rational and philosophical assumptions that underlie a particular study. The Methodology is an analysis of all the methods and procedures of the investigation. Methodologies are applied during the initial stage of the research process.

➤ **Method** – a (systematic ?) codified series of steps taken to complete a certain task or to reach a certain objective.

The “scientific method” attempts to minimize the influence of the researchers’ bias on the outcome of an experiment. Methods are just behaviors or tools used to select a research technique. Methods are applied during the later stage of the research study.

Research Methodology	vs	Research Method
Refers to a process for effectively addressing research problems		Describes the methods that the researcher uses to conduct the research
Investigate the various methods that can be used to conduct experiments, tests, surveys		It involves conducting an experiment, test, survey
Used to find a solution to the research problem		Used to achieve the research objective
Aims to answer this question. How did the researcher complete his study?		Aims to answer this question. What tools did the researcher use to complete his research?
Any research must start with research methodology because it provides the way for choosing the most suitable research methods		Research, whether it be scientific or not, must have research methods

➤ Definition of Research?

- **Research is a scientific way of answering questions and testing hypotheses.**
- **Research is the systematic and objective analysis and recording of controlled observations that may lead to the development of generalizations, principles, or theories, resulting in prediction and possible control of events.**

➤ What is Scientific Research?

- **Scientific research is the systematic investigation of scientific theories and hypotheses (The scientific method is a process for experimentation that is used to explore observations and answer questions).**
- **Scientific research may be defined as a systematic, controlled, empirical, and critical investigation of hypothetical propositions about the presumed relations among observed phenomena.**

➤ Scientific Research

- **A process of rigorous reasoning based on interactions among theories methods, and findings.**
- **Builds on understanding derived from the objective testing of models or theories.**
- **Accumulation of scientific knowledge is laborious, plodding, circuitous, and indirect.**
- **Scientific knowledge is developed and improved through critique of contested findings, replication, and convergence.**
- **Scientific knowledge is developed through sustained efforts.**
- **Scientific inquiry must be guided by fundamental principles.**

➤ Objectives of Scientific Research

1. **Highlighting a fact or developing a solution to a problem.**
2. **Completing previous research and clarifying what is ambiguous.**
3. **Coming up with a new idea or invention.**
4. **Collecting miscellaneous information.**
5. **Re-presenting topics in a new way.**
6. **Developing society and spreading awareness.**
7. **Describe, explain, predict, and solve problems.**

➤ Characteristics of good research

1. The title is clear, comprehensive and attractive.
2. The research steps and objectives must be specific.
3. Sufficient knowledge of the research topic.
4. Interconnection between parts of the research.
5. Clarity of the research method and all information.
6. Contributing to increasing knowledge.
7. Scientific honesty in communicating information.
8. Availability of information sources.
9. Objectivity and applicability.

➤ Characteristics of a good researcher

1. **Very knowledgeable.**
2. **Ability to choose good topics.**
3. **The ability to innovate.**
4. **The ability to organize, express and achieve.**
5. **Patience, lack of despair and endurance.**
6. **Desire and focus.**
7. **Impartiality and neutrality.**
8. **Honesty and objectivity.**
9. **Love research and good listening.**

➤ Basic Elements of the Scientific Methods

The scientific method refers to a standardized set of techniques for building scientific knowledge, such as how to make valid observations, how to interpret results, and how to generalize those results.

Replicability:

- Replicate or repeat a scientific study and obtain similar, if not identical, results.

Precision:

- Theoretical concepts that are to be defined so that others can use those definitions to measure those concepts and test that theory.

Falsifiability:

- Theories that cannot be tested or falsified are not scientific theories and any such knowledge is not scientific.

Parsimony:

- When there are multiple explanations of a phenomenon, scientists must always accept the simplest or logically most economical explanation.

➤ CHARACTERISTICS OF SCIENTIFIC METHODS

- They are orderly & systematic processes.
- Scientists attempt to control external factors that are not under direct investigation.
- Their findings are based on the empirical evidence.
- Findings of scientific methods can be generalized, which means that they can be used in situations other than the one under study.
- Scientific methods are based on assumptions or hypotheses.
- They are basically conducted to develop or test hypotheses.

➤ LIMITATIONS OF SCIENTIFIC METHODS

1

- **The moral or ethical problem**

2

- **Human complexity**

3

- **Measurement problems**

4

- **External variable control problems.**

➤ 1- Moral or Ethical Problem

The constraint of involving humans, which is essential for observing human behavior for experimentation, is one such ethical problem.

➤ 2- Human complexity

- **Human behavior is complicated, subtle & varied.**
- **It is very difficult to categorize human behavior.**
- **When human behavior is studied & analyzed by other human beings, personal biases come into the picture & distort the analytical facts.**

➤ 3- Measurement problems

- Different aspects of human behavior are psychological in nature, which cannot be accurately measured.
- Human behavior is not uniform, certain, or predictable.
- All the people do not behave in the same way in similar circumstances.
- Therefore, with humans as subjects, measurements become a challenge.

➤ 4- External variable control problems

Scientific studies conducted on humans may have a limitation of weak or no control over external variables in scientific activity.

➤ Researcher Competencies

☐ Researcher Scientific Competence

- A set of knowledge and skills necessary to carry out research activities.

☐ Researcher Logical Competence

- Logical researchers observe and analyze phenomena, reactions, and feedback and then draw conclusions based on that input. The researchers can justify their strategies, actions, and decisions based on the facts they gather.

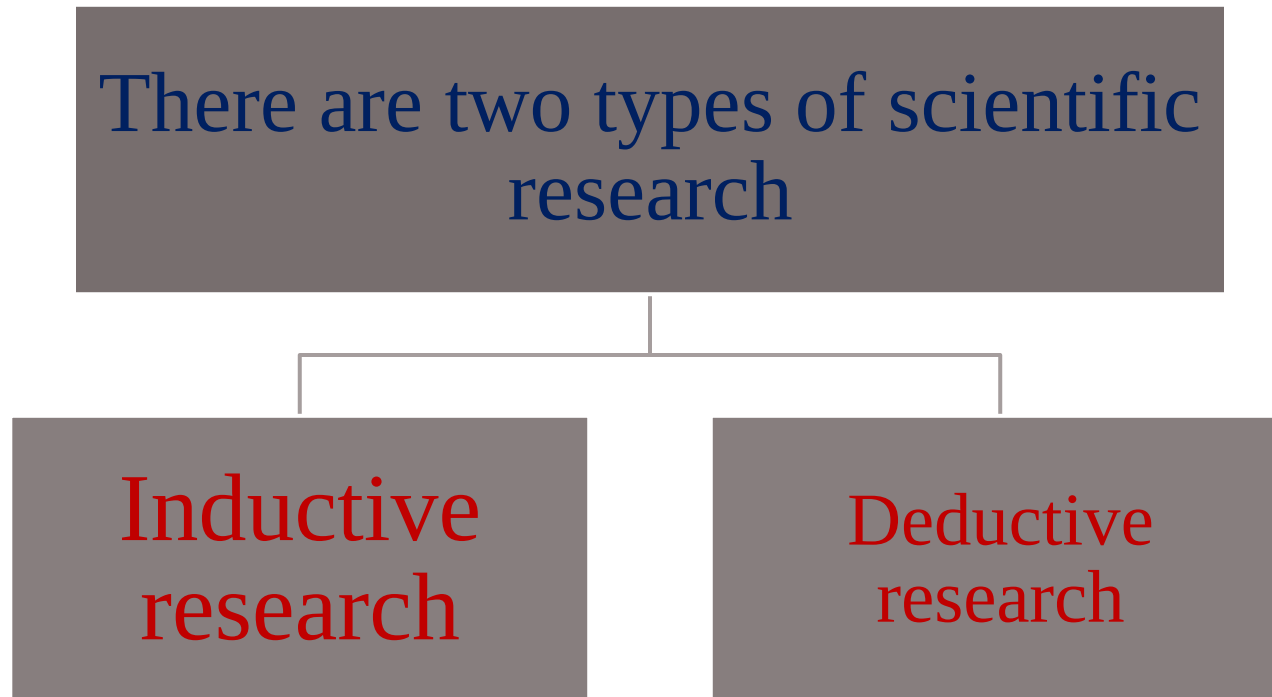
☐ Researcher Technical Competence

- The technical definition is having special and usually practical knowledge, especially of a mechanical or scientific subject. Technical definition mostly refers to the aspect of explaining or describing any technical terms or terminologies.

☐ Researcher Ethical Competence

- Ethical competence is often described as an experience acquired through the combination of knowledge and practice.
- Incorporating ethical principles into research practice.
- May involve a balance between and within principles and practices.
- All stages, all those involved, from inception of research through to completion and publication of results and beyond.
- Scientific honesty in presenting the pros and cons of the results.

➤ Types of Scientific Research



Inductive Research:

- It is theory-building research. Researchers assume theoretical concepts and patterns from observed data (from a specific case to the general principle).

Deductive Research:

- It is theory-testing research. The researcher tests concepts and patterns using empirical data (from the general case to the specific).

On the basis of purpose, there are 3 more types of scientific research:

- **Exploratory.**
- **Descriptive.**
- **Explanatory.**

Exploratory:

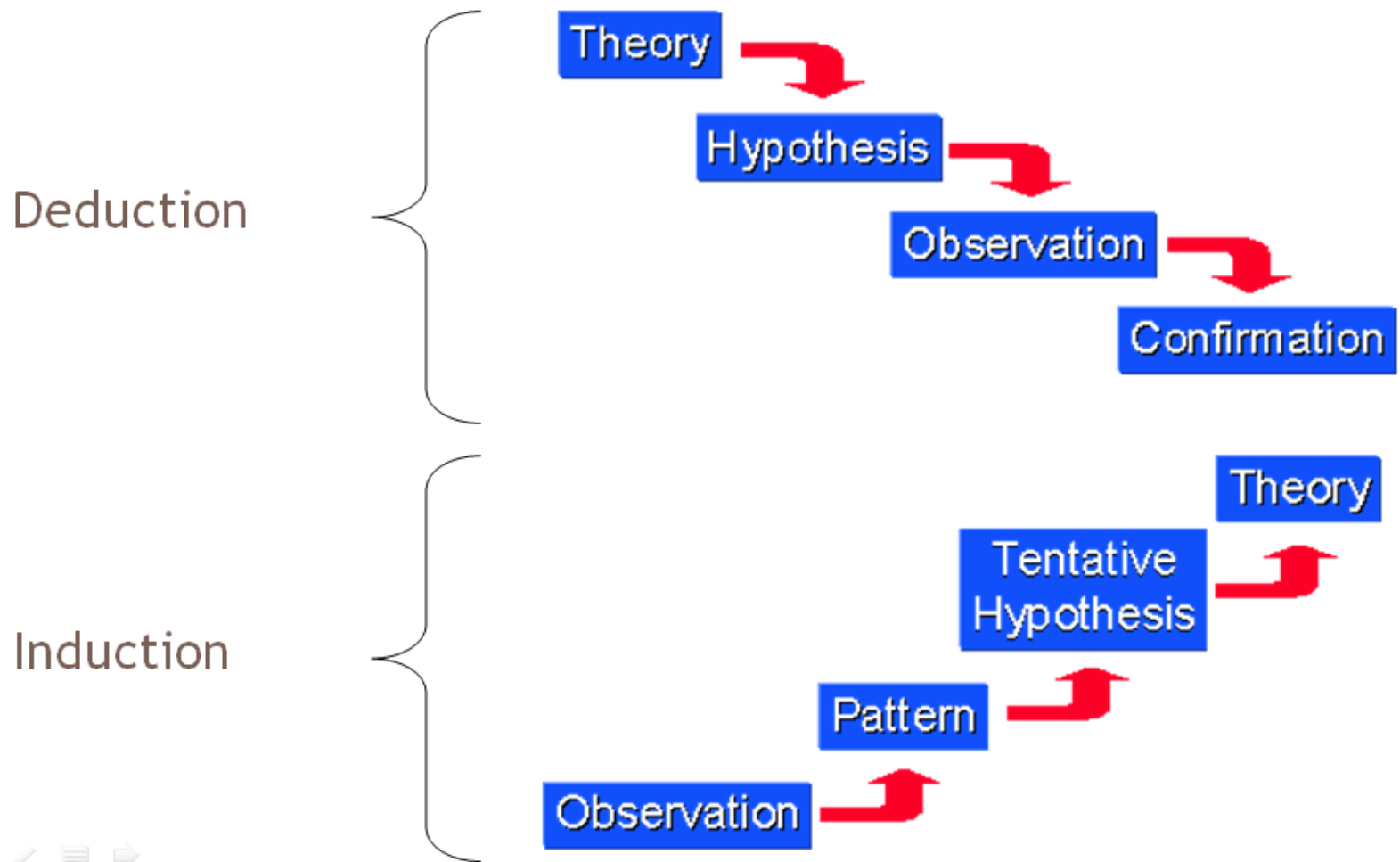
- Exploratory research is conducted in new areas of interest or inquiry.

Descriptive:

- It is careful observation and detailed documentation of a phenomena based on scientific methods therefore these observations are more reliable than casual observation example of this research are Tabulation of demographic statistics by the census department.

Explanation:

- This research seeks the explanation of the observed phenomena, problems, or behavior. Description research examines the what and where of phenomena while explanatory research answers why and how types of questions.



➤ Classification of Research on The Basis of Inquiry Mode

Classification based on Inquiry Mode: a) Structured approach, b) Unstructured approach. The structured approach to enquiry is usually classified as quantitative research and unstructured as qualitative research.

Qualitative Research:

- Is research undertaken to gain insights concerning attitudes, beliefs, motivations, and behaviors of individuals, to explore a social or human problem.

Quantitative Research:

- Is research concerned with the measurement of attitudes, behaviors, and perceptions. It includes interviewing methods such as telephone, intercept, and door-to-door interviews, as well as self-completion methods such as mailouts and online surveys.